

Task-Specific and Uncertainty-Aware Image Quality Assessment for Prostate MRI

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Abstract. Image quality in prostate MRI is usually assessed visually, yet the clinical value of a scan ultimately depends on whether it supports reliable downstream decisions. This project develops a task-specific and uncertainty-aware framework for prostate MRI image quality assessment. The framework uses T2-weighted imaging, high-b-value diffusion-weighted imaging, apparent diffusion coefficient maps, prostate masks, and task labels to train downstream models for lesion segmentation, Gleason grade group ≥ 2 classification, and PI-RADS ≥ 4 classification. Out-of-fold downstream predictions are converted into ternary image-quality labels: accept, caution, and reject, which supervise a 3D DenseNet IQA model. Reliability is assessed using softmax confidence, predictive entropy, and feature-space novelty estimated with Mahalanobis distance. Experiments use patient-level splitting, locked hold-out testing, and preprocessing controls designed to prevent leakage. The main evaluation tests whether predicted IQA categories, uncertainty, and novelty are associated with downstream task performance. The proposed approach aims to move prostate MRI IQA from subjective visual scoring toward objective, task-aware decisions that identify both unsuitable scans and predictions requiring caution.

Keywords: Prostate MRI; image quality assessment; task amenability; uncertainty; novelty detection; deep learning.

Revision note: yellow highlighting marks edited or newly added wording in this version. Removed placeholder text is represented by the replacement wording shown here.

1 Introduction

Prostate magnetic resonance imaging (MRI) is central to prostate assessment, treatment planning, and cancer diagnosis. Standardized frameworks such as PI-RADS have improved the consistency of prostate MRI acquisition, quality review, and clinical interpretation (Turbey et al., 2019). In both multi-parametric and bi-parametric protocols, T2-weighted imaging (T2WI) provides anatomical context, whereas diffusion-weighted imaging (DWI) provides functional information that is important for detecting clinically significant prostate cancer. However, prostate DWI is vulnerable to artefacts caused by rectal gas, patient motion, signal loss, and geometric distortion (Tarroni et al., 2019). These artefacts can misalign DWI with T2WI, impair lesion localization, and reduce the reliability of downstream clinical decisions (Donato et al., 2014).

Image quality is therefore not only a visual property, but also a question of task reliability. A scan that appears visually acceptable may still be unreliable for Gleason grade group classification, PI-RADS classification, or lesion segmentation. Conversely, a scan with visible artefacts may remain adequate for a specific clinical task. Saeed et al. formalized this idea as task amenability: the degree to which an image supports a given downstream task. In their framework, a downstream task model generates task performance information, while an IQA controller learns from task outcomes rather than from purely visual labels (Saeed et al., 2022). Related work in ultrasound has also shown that acquisition quality can be evaluated through downstream model performance, for example when estimating fetal ultrasound operator skill from task-specific data quality (Wang et al., 2022).

Standardized systems such as PI-QUAL have improved clinical decision-making and prostate MRI quality reporting (Giganti et al., 2020; Karanasios et al., 2022). Deep learning methods have also shown promise for binary and multi-parametric prostate MRI quality assessment (Cipollari et al., 2022; Alis et al., 2023). Together, these studies suggest that automated prostate MRI IQA can be developed, but important technical gaps remain.

First, most existing prostate MRI IQA models are trained against visual quality labels rather than task-derived supervision. Second, task amenability has not yet been implemented as a complete prostate MRI pipeline in which downstream clinical tasks define image-quality labels. Third, many approaches assess image quality and prediction reliability separately, even though clinical use requires both pieces of information at the same time.

This work develops a task-specific and uncertainty-aware prostate MRI IQA framework. The proposed pipeline uses T2WI, high-b-value DWI, ADC maps, prostate masks, and where available lesion masks as inputs, and assigns each case to one of three clinically actionable categories: accept, caution, or reject. Image quality is defined with respect to downstream clinical tasks, including lesion segmentation, Gleason grade group ≥ 2 classification, and PI-RADS ≥ 4 classification. Cases that support accurate and confident downstream predictions are treated as higher-quality examples for task-specific IQA supervision. The framework also incorporates predictive uncertainty and feature-space novelty so that scans unlike the training distribution can be flagged even when their predicted quality category appears acceptable. The goal is to move prostate MRI IQA from subjective visual scoring toward objective, task-aware decisions that account for both scan suitability and prediction reliability.

2 Methods

2.1 Overview

We propose a task-specific and uncertainty-aware prostate MRI image-quality assessment (IQA) framework. The aim is to identify cases that can support reliable downstream clinical decisions. The framework contains five stages: (i) patient-level data organization and preprocessing, (ii) downstream task model training, (iii) task-derived quality-label generation, (iv) ternary IQA model training, and (v) uncertainty and novelty evaluation. The final IQA output assigns each case to one of three clinically actionable categories: accept, caution, or reject.

2.2 IQA model architecture

The main IQA model is a 3D DenseNet-based classifier. The input consists of high-b-value DWI, ADC, T2WI, and the prostate mask, with lesion masks used only where required by the downstream task. The default multimodal input is DWI + ADC + T2WI + prostate mask. The model outputs three ordered classes: reject, caution, and accept. Training uses cross-entropy combined with an expected-score mean absolute error term so that errors between adjacent quality categories are penalized less strongly than errors between distant categories.

Let p_k be the softmax probability for ordered class k in $\{0, 1, 2\}$, where 0 denotes reject, 1 denotes caution, and 2 denotes accept. The expected quality score is defined as $s = \sum_k k p_k$. The ordinal loss is $L_{ord} = L_{CE} + \lambda |s - y|$, where y is the true ordered label and λ controls the ordinal penalty. This loss preserves the categorical classification objective while additionally penalizing predictions according to the distance between the expected quality score and the true ordered label.

Class imbalance is handled using inverse-frequency weighted sampling within the training set. Data augmentation is applied to MRI channels using spatial and intensity transformations, and masks are transformed consistently with their corresponding images. The best checkpoint is selected on the validation set using macro-F1, AUC, or validation loss, depending on the experiment. All model settings are fixed before evaluation on the locked hold-out test set.

2.3 Uncertainty and novelty estimation

Each IQA prediction is accompanied by uncertainty and novelty estimates. Predictive uncertainty is quantified using softmax confidence and entropy; low confidence or high entropy indicates that the model is uncertain between quality categories. Feature-space novelty is estimated from embeddings extracted from the penultimate layer of the trained model. Training-set embeddings are used to fit a regularized multivariate Gaussian distribution. For each validation or test case, Mahalanobis distance is computed between its feature vector and the training feature distribution.

Novelty and uncertainty are not treated as image-quality labels themselves. Instead, they are interpreted as reliability indicators. A scan may be assigned to the accept category while still being unfamiliar to the model, and such cases should be flagged for caution. The final triage decision therefore combines the predicted IQA category with uncertainty and novelty scores.

3 Experiments

3.1 Dataset and inclusion criteria

The study uses a retrospective prostate MRI cohort containing pre-treatment T2WI, high-b-value DWI, and, where available, ADC maps. The lesion-segmentation task requires T2WI, high-b-value DWI, and lesion masks. The classification tasks require patient-level Gleason grade group labels and radiologist-assigned PI-RADS labels. Each case is represented in a master manifest containing the patient identifier, scan identifier, paths to T2WI, DWI, ADC, prostate mask, lesion mask, available quality labels, downstream clinical labels, acquisition metadata, and split assignment.

Clinical binary labels are defined as Gleason grade group ≥ 2 versus < 2 and PI-RADS ≥ 4 versus < 4 . When multiple DWI acquisitions are available, the high-b-value volume is selected, with $b \geq 1400 \text{ s/mm}^2$ used as the preferred criterion. Cases with ambiguous or missing required inputs are excluded only from the specific analysis that requires those inputs.

3.2 Patient-level data splitting and leakage control

All splitting is performed at the patient level to prevent data leakage. All images and labels from the same patient are assigned to the same subset. The dataset is divided into training, validation, and locked hold-out test subsets using a stratified patient-level split that preserves the distribution of the main image-quality label where available. The hold-out test set is not used for training, model selection, hyperparameter tuning, preprocessing optimization, or task-derived label generation. Patient identifiers, scan identifiers, random seeds, split assignments, and preprocessing parameters are saved in locked split files to support reproducibility.

3.3 Image preprocessing and visual quality control

All images are preprocessed using a fixed pipeline. T2WI, high-b-value DWI, ADC maps, prostate masks, and lesion masks are loaded for each eligible case. DWI and ADC are aligned to T2WI. T2WI, DWI, ADC, and masks are resampled to a common voxel spacing; linear interpolation is used for image volumes and nearest-neighbor interpolation is used for masks. A prostate-centered crop is extracted using either the prostate-mask center or the image center, depending on mask availability and task requirements. Intensities are normalized using a fixed non-zero percentile-based method, and final volumes are cropped or padded to a common region-of-interest size.

Visual quality control is performed before model training. For each case, axial contact sheets, triplanar views, T2WI-DWI side-by-side views, and mask overlays are generated. Metadata including image shape, voxel spacing, orientation, and intensity percentiles are saved to a summary table.

3.4 Downstream clinical tasks

Three downstream clinical tasks are considered: lesion segmentation, Gleason grade group classification, and PI-RADS classification. The lesion-segmentation model uses multimodal MRI inputs to predict lesion masks and is evaluated using Dice similarity coefficient, lesion-wise sensitivity, and boundary-distance metrics. Patient-level Gleason classification is formulated as Gleason grade group ≥ 2 versus < 2 . PI-RADS classification is formulated as PI-RADS ≥ 4 versus < 4 . Both classification models use multimodal MRI inputs and output the probability of the positive class. Classification performance is evaluated using AUC, accuracy, sensitivity, and specificity.

Downstream task models serve two purposes: they provide baseline clinical-task performance, and their predictions are used to derive task-specific image-quality labels for IQA supervision. To avoid optimistic label generation, task-derived quality labels in the development set are generated from patient-level out-of-fold downstream predictions rather than from predictions on the same data used to train the downstream model. For segmentation, out-of-fold predictions are converted into ternary task-quality labels using lesion-detection and overlap thresholds. For classification, out-of-fold predictions are converted into ternary labels using correctness and confidence thresholds. Thresholds are selected on the validation set and fixed before final testing. A case is considered high task-specific quality if it supports accurate and confident downstream prediction. When multiple downstream tasks are available for the same case, task-derived labels are analyzed separately for each task and may also be combined into a composite task-specific quality label. This allows evaluation of whether a scan is generally acceptable or specifically suitable for one clinical task but not another.

3.5 Task-specific evaluation

The primary analysis tests whether predicted image quality is associated with downstream performance. Cases in the locked test set are grouped by predicted IQA category. Downstream segmentation and classification performance are then compared across predicted quality groups. We further evaluate whether uncertainty and novelty scores predict downstream failure. For segmentation, failure is defined as missed lesion detection or low overlap with the reference mask. For classification, failure is defined as incorrect prediction or low-confidence prediction. Associations between IQA score, uncertainty, novelty distance, and downstream performance are estimated using correlation analysis, regression models, and failure-prediction AUC.

The expected behavior is that accept cases should show better downstream segmentation and classification performance than caution or reject cases. High-novelty and high-uncertainty cases are expected to have higher error rates, lower segmentation overlap, or larger boundary errors. This analysis directly tests whether the IQA model captures task amenability rather than only visual image quality.

3.6 Statistical analysis

All final statistical analyses are performed on the locked hold-out test set. For classification endpoints, AUC, accuracy, sensitivity, and specificity are reported with 95 percent confidence intervals. AUC differences are compared using DeLong tests where applicable. For segmentation endpoints, Dice score, lesion-wise sensitivity, and boundary-distance metrics are summarized by IQA category. Associations between IQA score, uncertainty, novelty, and downstream performance are assessed using Spearman correlation, regression analysis, and failure-prediction AUC.

3.7 Implementation and reproducibility

All experiments are implemented in Python using PyTorch and MONAI for model training and SimpleITK for preprocessing. The pipeline saves the master manifest, split files, preprocessing manifest, random seeds, model checkpoints, prediction tables, novelty scores, and analysis outputs. The locked test split is protected from accidental use in training scripts. Model evaluation scripts export case-level predictions to support transparent error analysis, and report-ready figures, including confusion matrices and novelty-versus-failure plots, are generated automatically.

4 Results

To be completed after model training, locked test-set evaluation, and statistical analysis are finalized.

5 Discussion

To be completed after the final results are available. This section should interpret whether task-specific IQA improves clinical reliability beyond visual quality assessment and should discuss limitations such as cohort size, label noise, scanner heterogeneity, and external validation.

6 Conclusion

To be completed after the final analysis. The conclusion should briefly state whether the proposed framework can identify scans that are suitable for downstream prostate MRI tasks while flagging uncertain or novel cases.

References

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